1 Law: slope only

($h := \bar{w} - \bar{w}_s$, $\bar{w}_s$ = visible surface, given; p fixed; c = the one constant; used on $h \geq 1.1$)

$$
\begin{aligned}
\bar{a}'(h;\, c) &= \frac{1-c}{h^p} \\
\frac{\partial \bar{a}'}{\partial c} &= -\frac{1}{h^p} \\
\frac{\partial \bar{a}'}{\partial h} &= -\frac{p\,\bar{a}'}{h} \\
\frac{\partial \bar{a}'}{\partial \bar{a}_w} &= 0
\end{aligned}
$$

2 State

$$\mathbf{x} = \left[\, \delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ c \,\right]^\top$$

3 True dynamics

($d = 2$; $\bar{a}'_{\text{true}}$ = the plant's slope; $r' := \bar{a}'_{\text{true}}(h) - \bar{a}'(h;\, c)$ the part of the slope the family cannot hold)

$$
\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k] \\
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k] \\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\bar{a}_w[k+1] &= \bar{a}_w[k] + \bar{a}'_{\text{true}}\big(\bar{w}[k] - \bar{w}_s\big)\,\Delta\bar{w}[k] \\
c[k+1] &= c[k] \\
\Delta\bar{w}[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k]
\end{aligned}
$$

4 Estimator

($\hat{\bar{w}}[k] = \bar{w}_d[k] - \delta\hat{\bar{w}}_3[k]$; $\hat{h}[k] = \hat{\bar{w}}[k] - \bar{w}_s$; $M[k] := \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k]$)

$$
\begin{aligned}
\bar{a}'[k] &= \frac{1-\hat{c}[k]}{\hat{h}[k]^p} \\
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] \\
\hat{\bar{a}}_w[k+1] &= \hat{\bar{a}}_w[k] + \bar{a}'[k]\,M[k] \\
\hat{c}[k+1] &= \hat{c}[k]
\end{aligned}
$$

5 Error dynamics

($e = \text{true} - \text{estimate}$; first order at the estimate; $\Delta\bar{w} - M = (1-\lambda_c)\, e_{\delta\bar{w}_3} + F_{dw}\, e_{\bar{a}_w} + \varepsilon_w$; $\bar{w} - \hat{\bar{w}} = -e_{\delta\bar{w}_3}$)

$$\bar{a}'_{\text{true}} - \bar{a}'[k] = \frac{p\,\bar{a}'[k]}{\hat{h}[k]}\, e_{\delta\bar{w}_3}[k] - \frac{1}{\hat{h}[k]^p}\, e_c[k] + r'[k]$$

$$\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k] \\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k] \\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c\, e_{\delta\bar{w}_3}[k] - F_{dw}[k]\, e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
e_{\bar{a}_w}[k+1] &= \Big(\bar{a}'[k](1-\lambda_c) + \frac{p\,\bar{a}'[k]}{\hat{h}[k]} M[k]\Big) e_{\delta\bar{w}_3}[k] + \big(1 + \bar{a}'[k] F_{dw}[k]\big) e_{\bar{a}_w}[k] \\
&\quad - \frac{M[k]}{\hat{h}[k]^p}\, e_c[k] + \bar{a}'[k]\, \varepsilon_w[k] + M[k]\, r'[k] \\
e_c[k+1] &= e_c[k]
\end{aligned}$$

$$F_e[k] = \begin{bmatrix}
0 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 \\
0 & 0 & \lambda_c & -F_{dw}[k] & 0 \\
0 & 0 & \bar{a}'[k](1-\lambda_c) + \dfrac{p\,\bar{a}'[k]}{\hat{h}[k]} M[k] & 1 + \bar{a}'[k] F_{dw}[k] & -\dfrac{M[k]}{\hat{h}[k]^p} \\
0 & 0 & 0 & 0 & 1
\end{bmatrix}$$

6 Error dynamics, row 4 expanded

(all quantities at step k; e = true − estimate; $h = \hat{h} - e_{\delta\bar{w}_3}$; $c = \hat{c} + e_c$)

(a) the two increments and their difference

$$\bar{a}_w[k+1] = \bar{a}_w[k] + \bar{a}'_{\text{true}}\,\Delta\bar{w}$$
$$\hat{\bar{a}}_w[k+1] = \hat{\bar{a}}_w[k] + \bar{a}'\,M$$
$$e_{\bar{a}_w}[k+1] = e_{\bar{a}_w}[k] + \bar{a}'_{\text{true}}\,\Delta\bar{w} - \bar{a}'\,M$$

(b) the two factors, each written as estimate plus its error

$$\begin{aligned}
\Delta\bar{w} - M &= \left[\Delta\bar{w}_d + (1-\lambda_c)\,\delta\bar{w}_3 + F_{dw}\,e_{\bar{a}_w} + \varepsilon_w\right] - \left[\Delta\bar{w}_d + (1-\lambda_c)\,\delta\hat{\bar{w}}_3\right] \\
&= (1-\lambda_c)\,e_{\delta\bar{w}_3} + F_{dw}\,e_{\bar{a}_w} + \varepsilon_w \\
\bar{a}'_{\text{true}}(h) &= \bar{a}'(h;\,c) + r' \\
\bar{a}'(h;\,c) - \bar{a}'(\hat{h};\,\hat{c}) &= \frac{\partial\bar{a}'}{\partial h}\,(h-\hat{h}) + \frac{\partial\bar{a}'}{\partial c}\,(c-\hat{c}) \qquad \text{first order} \\
&= \Big(-\frac{p\,\bar{a}'}{\hat{h}}\Big)(-e_{\delta\bar{w}_3}) + \Big(-\frac{1}{\hat{h}^p}\Big)(e_c) \\
\bar{a}'_{\text{true}} - \bar{a}' &= \frac{p\,\bar{a}'}{\hat{h}}\,e_{\delta\bar{w}_3} - \frac{1}{\hat{h}^p}\,e_c + r'
\end{aligned}$$

(c) the product difference, exact

$$\begin{aligned}
\bar{a}'_{\text{true}}\,\Delta\bar{w} - \bar{a}'\,M &= \bar{a}'_{\text{true}}\,\Delta\bar{w} - \bar{a}'\,\Delta\bar{w} + \bar{a}'\,\Delta\bar{w} - \bar{a}'\,M \qquad \text{add and subtract } \bar{a}'\,\Delta\bar{w} \\
&= (\bar{a}'_{\text{true}} - \bar{a}')\,\Delta\bar{w} + \bar{a}'\,(\Delta\bar{w} - M) \\
&= (\bar{a}'_{\text{true}} - \bar{a}')\,M + (\bar{a}'_{\text{true}} - \bar{a}')(\Delta\bar{w} - M) + \bar{a}'\,(\Delta\bar{w} - M) \qquad \Delta\bar{w} = M + (\Delta\bar{w} - M)
\end{aligned}$$

(d) every term after substituting (b)

$$
\begin{aligned}
(\bar{a}'_{\text{true}} - \bar{a}')\, M &= \frac{p\,\bar{a}'}{\hat{h}}\, M\, e_{\delta\bar{w}_3} && (1)\\
&\quad - \frac{M}{\hat{h}^p}\, e_c && (2)\\
&\quad + M\, r' && (3)\\
\bar{a}'\,(\Delta\bar{w} - M) &= \bar{a}'\,(1-\lambda_c)\, e_{\delta\bar{w}_3} && (4)\\
&\quad + \bar{a}'\, F_{dw}\, e_{\bar{a}_w} && (5)\\
&\quad + \bar{a}'\, \varepsilon_w && (6)\\
(\bar{a}'_{\text{true}} - \bar{a}')(\Delta\bar{w} - M) &= \frac{p\,\bar{a}'}{\hat{h}}\,(1-\lambda_c)\, e_{\delta\bar{w}_3}\, e_{\delta\bar{w}_3} && (7)\ \text{second order}\\
&\quad + \frac{p\,\bar{a}'}{\hat{h}}\, F_{dw}\, e_{\delta\bar{w}_3}\, e_{\bar{a}_w} && (8)\ \text{second order}\\
&\quad + \frac{p\,\bar{a}'}{\hat{h}}\, e_{\delta\bar{w}_3}\, \varepsilon_w && (9)\ \text{second order}\\
&\quad - \frac{1-\lambda_c}{\hat{h}^p}\, e_c\, e_{\delta\bar{w}_3} && (10)\ \text{second order}\\
&\quad - \frac{F_{dw}}{\hat{h}^p}\, e_c\, e_{\bar{a}_w} && (11)\ \text{second order}\\
&\quad - \frac{1}{\hat{h}^p}\, e_c\, \varepsilon_w && (12)\ \text{second order}\\
&\quad + (1-\lambda_c)\, r'\, e_{\delta\bar{w}_3} && (13)\ \text{second order}\\
&\quad + F_{dw}\, r'\, e_{\bar{a}_w} && (14)\ \text{second order}\\
&\quad + r'\, \varepsilon_w && (15)\ \text{second order}
\end{aligned}
$$

(e) first-order terms collected by error

$$
\begin{aligned}
e_{\bar{a}_w}[k+1] &= e_{\bar{a}_w}[k] + (1) + (4) + (5) + (2) + (6) + (3)\\
&= \Big(\bar{a}'\,(1-\lambda_c) + \frac{p\,\bar{a}'}{\hat{h}}\, M\Big)\, e_{\delta\bar{w}_3} && (4)+(1) && \to F_e(4,3)\\
&\quad + \big(1 + \bar{a}'\, F_{dw}\big)\, e_{\bar{a}_w} && 1+(5) && \to F_e(4,4)\\
&\quad - \frac{M}{\hat{h}^p}\, e_c && (2) && \to F_e(4,5)\\
&\quad + \bar{a}'\, \varepsilon_w && (6) && \to Q,\ \mathbf{g}_4 = \bar{a}'\\
&\quad + M\, r' && (3) && \to \text{not in } F_e,\ \text{not in } Q
\end{aligned}
$$

7 Measurement

$(\nabla_d \bar{w}_d[k] := \bar{w}_d[k] - \bar{w}_d[k-d])$

$$
\begin{aligned}
y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k-d] + n_a[k] \\
\hat{y}_1[k] &= \delta\hat{\bar{w}}_1[k] \\
\hat{y}_2[k] &= \hat{\bar{a}}_w[k] - \bar{a}'[k]\,\nabla_d \bar{w}_d[k] \\
H[k] &= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & -\dfrac{p\,\bar{a}'[k]}{\hat{h}[k]}\nabla_d\bar{w}_d[k] & 1 & \dfrac{\nabla_d \bar{w}_d[k]}{\hat{h}[k]^p} \end{bmatrix}
\end{aligned}
$$

8 Noise

$$
\begin{aligned}
Q[k] &= Q_{33}[k]\,\mathbf{g}\mathbf{g}^\top, \qquad \mathbf{g} = \begin{bmatrix} 0,\ 0,\ -1,\ \bar{a}'[k],\ 0 \end{bmatrix}^\top, \qquad Q_{55} = 0 \\
R[k] &= \operatorname{diag}\big(R_1,\ R_2[k]\big)
\end{aligned}
$$

Seed and prior (plane init)

$$
\begin{aligned}
p &= 2.042 \\
\hat{c}[0] &= c_{\text{plane}} = +0.037 \\
\hat{\bar{a}}_w[0] &= D_\perp\Big(\frac{1}{\bar{w}[0] - \bar{w}_s}\Big) \\
P[0] &= \operatorname{diag}\big(P_{11}[0],\ P_{22}[0],\ P_{33}[0],\ P_{44}[0],\ \sigma_{c,0}^2\big), \qquad \sigma_{c,0} = 0.5 \\
P_{44}[0] &= \bar{a}'\big(\bar{w}[0] - \bar{w}_s\big)^2\,\sigma_{\bar{w}_s}^2 + \sigma_{a,0}^2
\end{aligned}
$$